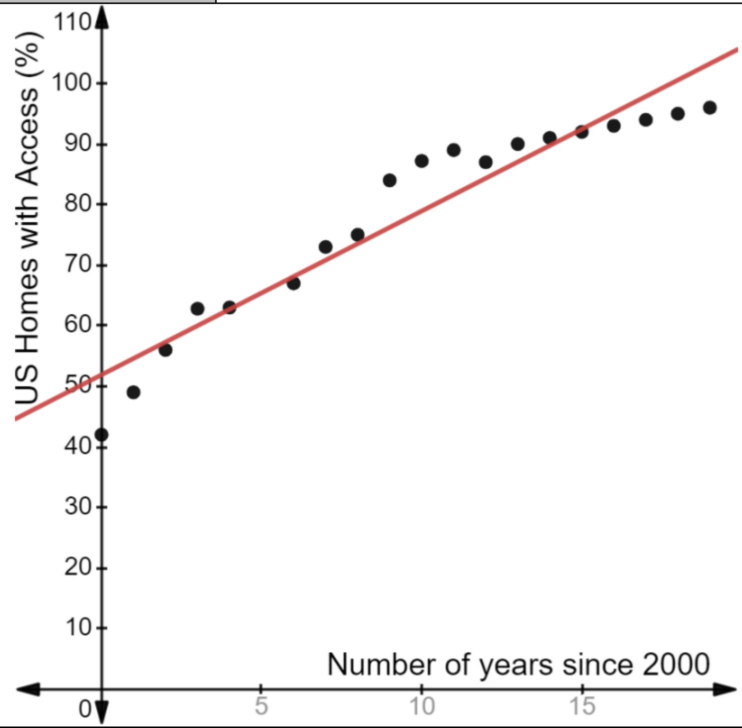
8th Grade Unit 11 Assessment Guide

General Information	Unit 11 begins building students' understanding of linear functions by examining relationships that have linear associations and contrasting these with relationships that are not linear. Students consider whether the data is best displayed using a line of fit or with a line graph. Having students relate the use of each to the purpose of the display may help students distinguish between scatter plots that show a trend as opposed to line graphs that show an analysis from one point to another. The draft test item specifications state that students will not be asked to identify a strong or weak association. The statistical definitions of these terms can vary by expert, so assessing these terms could result in a debate over whether a display qualifies. Within classroom instruction, these terms are used where the teacher can use the informal definitions and allow students to describe scatter plots using this terminology. The understanding built in this unit is an important foundation for students as they enter Algebra 1. Teachers will find that the assessment does not require the students to define variables or label graphs, yet these are important skills for students as they work with bivariate data. These were avoided because, at times, the variables could easily be defined in either manner so assessing this is futile. Having students collect data and to investigate as they did with Barbie Bungee is an excellent way to have students experience the variables. If there is time, teachers may look for other opportunities for students to experience the bivariate data in this way.
Calculator Information	The questions on the unit assessment are calculator neutral. Teachers can determine whether students should use a calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.8.DP.1.3
		**Recommended Scoring** **(4 Total Points)**	Consider using a ruler when grading student papers to ensure the trend line considers the trend of all of the data.
Question	**1b**	**Benchmark(s)**	MA.8.DP.1.3
The line of fit shows	○ a closer analysis from each point to the other in the dataset. ○ that the independent variable is the percentage of US households with access to cell phones. ○ the overall relationship between the number of years since 2000 and the percentage of US households with access to cell phones. ○ that there is no association between the number of years since 2000 and the percentage of US households with access to cell phones.	**Recommended Scoring** **(3 Total Points)**	Consider giving no partial credit.

Question	2	Benchmark(s)	MA.8.DP.1.2

	Characteristics				
Scatter Plot	**Has a linear association**	**Has a nonlinear association**	**Positive Association**	**Negative Association**	**Outlier(s)**
A	■	□	□	■	■
B	■	□	□	■	□
C	■	□	■	□	□
D	□	■	□	□	□
E	□	■	□	□	□
F	■	□	■	□	□

Recommended Scoring

(22 Total Points)

Consider 2 points for each correct selection.

Question	3	Benchmark(s)	MA.8.DP.1.3

Line
- ○ D
- ○ H
- ○ R

is the line of fit because
- ○ it goes through the origin.
- ○ it goes through exactly two points and cuts the data in half.
- ○ approximately half of the data is above the line and half of the data is below the line.

Recommended Scoring

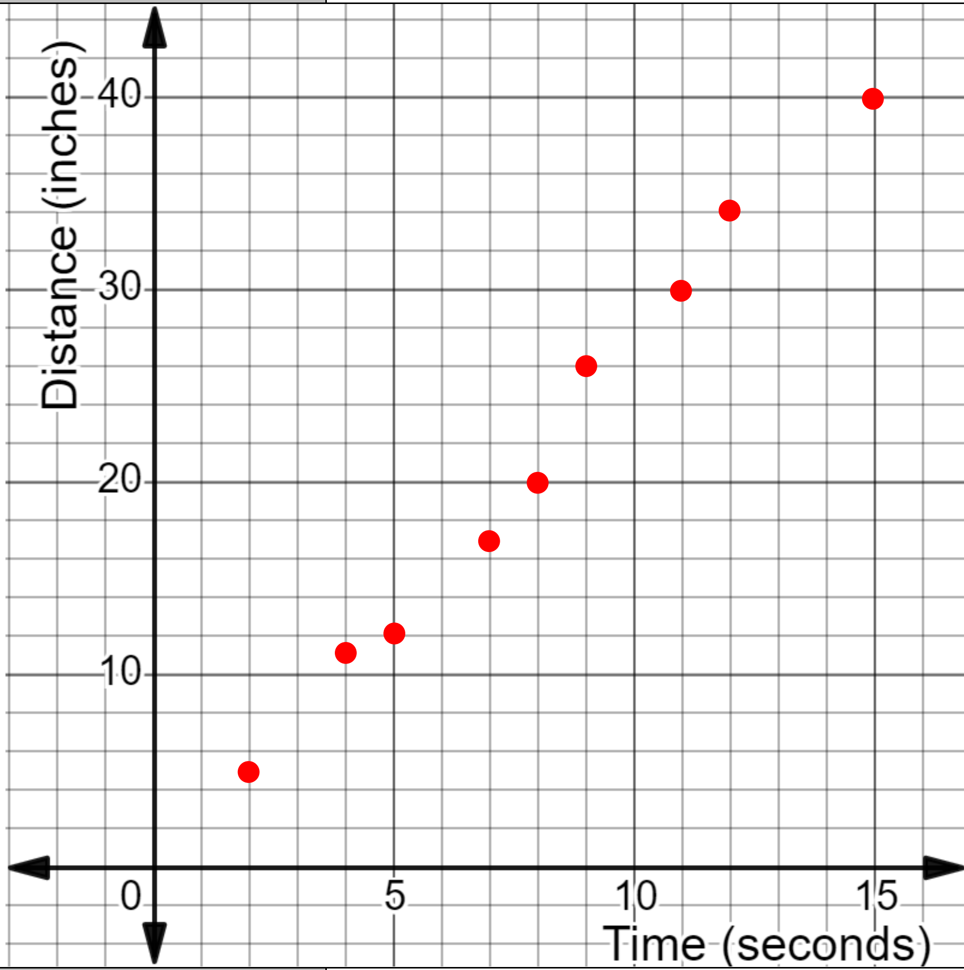
(4 Total Points)

Consider giving 2 points for each correct selection.

Question	4	Benchmark(s)	MA.8.DP.1.1
 		**Recommended Scoring** **(24 Total Points)**	Consider giving 2 points for each correctly graphed point and 8 points for creating a line graph.

Question	5a	Benchmark(s)	MA.8.DP.1.1
Complete the statements. The statistician should choose to create a [○ line graph / ○ scatter plot] where the [○ dependent / ○ independent] variable is the price of a box of citrus, in dollars. Graph [○ K / ○ N / ○ R / ○ T / ○ W / ○ Z] is the best choice for the statistician to use.		**Recommended Scoring** **(7 Total Points)**	Consider giving 2 points for each correct selection in first statement and 3 points for the correct selection in the second statement.

Question	5b	Benchmark(s)	MA.8.DP.1.1
Complete the statements. The accountant should choose to create a [○ line graph / ○ scatter plot] where the [○ dependent / ○ independent] variable is the price of a box of citrus, in dollars. Graph [○ K / ○ N / ○ R / ○ T / ○ W / ○ Z] is the best choice for the accountant to use.		**Recommended Scoring** **(7 Total Points)**	Consider giving 2 points for each correct selection in first statement and 3 points for the correct selection in the second statement.

Question	6a	Benchmark(s)	MA.8.DP.1.1
		**Recommended Scoring** **(18 Total Points)**	Consider giving 2 points for each correctly graphed point.

Question	6b	Benchmark(s)	MA.8.DP.1.2
The scatter plot shows that the data has	○ no association. ○ a nonlinear association. ○ a positive linear association. ○ a negative linear association.	**Recommended Scoring** **(3 Total Points)**	Consider giving no partial credit.

<table>
<tr><th>Question</th><th>7</th><th>Benchmark(s)</th><th>MA.8.AR.3.5</th></tr>
<tr><td colspan="2">The slope of the equation that fits the data

means that, on average, the [○ number of years / ○ average working hours] [○ increases / ○ decreases]

[○ 2 / ○ 21 / ○ 3,094] [○ years / ○ working hours] for every [○ 2 / ○ 21 / ○ 3,094] [○ years. / ○ working hours.]

The y-intercept means that in the year [○ 1870, / ○ 3094,] the average number

of working hours over a full year a worker in Italy worked was [○ 2. / ○ 21. / ○ 1,870. / ○ 3,094.]</td><td>Recommended Scoring

(8 Total Points)</td><td>Consider giving 4 points for each statement. Consider no partial credit for either statement. Students should complete all blanks in the statement correctly to demonstrate understanding of the interpretation.</td></tr>
</table>